MCO-1D ORNITHOPTER

Commissioning Manual — Revision 12

A biomimetic eight-wing metachronal cascade flapping-wing aircraft designed as a Search and Rescue mission testbed. This document covers every aspect of the MCO-1D from hardware assembly through software installation, bench testing, commissioning flights, and experimental discovery operations. It assumes no prior knowledge of the project.

Aircraft designation	MCO-1D
Configuration	Eight wings in four tandem pairs, metachronal cascade
Mission	Search and Rescue testbed, experimental biomimetics
MTOW	3.448 kg (7.60 lb)
Weight	33.8 N
Stall speed	4.85 m/s (9.4 knots) out-of-ground-effect
VNE	28.0 m/s (54.4 knots)
Frequency ceiling	3.0 Hz (operational limit)
Wing chord	165 mm
Propulsion	4× GoolRC 5010-14 360KV motors via ODrive CAN-FD
Control computer	Raspberry Pi 4 (4 GB recommended)
Software	Wave Library Python — 42,115 lines
Document version	Revision 12 — July 2026

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PART I — PROJECT OVERVIEW

SECTION 1 — WHAT THE MCO-1D IS

1.1 The Aircraft

The MCO-1D is a prototype ornithopter — an aircraft that flies by flapping its wings rather than by fixed wings or rotors. It has eight wings arranged in four tandem pairs from front to rear. The wings flap in a staggered sequence called a metachronal cascade, where each pair begins its stroke slightly after the pair in front. This is the same pattern used by plesiosaurs, dragonflies, and mantis shrimp — animals that evolved this locomotion independently in very different environments. The aircraft weighs 3.448 kilograms, has a stall speed of approximately 4.85 metres per second, and cannot stall in the conventional sense because it generates thrust and lift by active flapping rather than by maintaining airspeed over a fixed surface.

1.2 Why This Aircraft Exists

The MCO-1D is a testbed for a planned lineage of Search and Rescue aircraft. SAR missions in difficult terrain require an aircraft that can hover over a narrow ledge, descend vertically into a forest gap, operate in wind conditions that would ground a conventional drone, and carry a sensor payload while maintaining a stable platform. Fixed-wing aircraft cannot hover. Multirotor drones are efficient in calm air but their rigid-blade rotors are limited in gusty conditions. A flapping-wing aircraft with eight independent wings has fundamentally different characteristics: it can reconfigure how its wings move to adapt to conditions in real time, and it has eight separate sources of lift and control authority where a quadrotor has only four.

1.3 What Makes This Design Different

Every wing on the MCO-1D has five independent control parameters: flapping amplitude (how far it moves), flapping frequency (how fast), angle of attack (the pitch of the wing surface), sweep angle (fore-aft position), and dihedral (up-down tilt). With eight wings this gives forty independent parameters that can be varied in flight every 20 milliseconds. No conventional aircraft has this degree of aerodynamic programmability. The software that manages these parameters is called the Wave Library, and its central feature is an optimizer that continuously discovers which parameter combinations work best for each flight condition.

1.4 Where the Project Stands

The MCO-1D is a first prototype. The software is complete and has been audited extensively but the aircraft has not yet flown. Three commissioning tests must be completed in sequence before it is considered flight-ready. This document explains every step from first assembly to experimental discovery flights. The person reading this document is expected to know nothing about the project — every term is defined when first used.

SECTION 2 — HOW THIS DOCUMENT IS ORGANISED

This manual is divided into nine parts. Read them in order for the first build. After that, each part stands alone for reference.

Part I — Overview	What the MCO-1D is and why it exists. The context for everything else.
Part II — Hardware	Every physical component described. What it is, where it goes, how to connect it. No prior electronics knowledge assumed.
Part III — Software	Installing the operating system, the Wave Library, and the ground station. Step by step from a blank SD card.
Part IV — Flight Control	How the software thinks. The optimizer, the hypothesis library, the fault tolerance system. Conceptual understanding before flying.
Part V — Commissioning	Three test sequences: bench test, hover test, forward flight. What to do, what to measure, what must pass before proceeding.
Part VI — Flight Operations	Normal procedures for every type of flight. Pre-flight checklist, normal flight, emergencies, backcountry, water operations.
Part VII — Discovery	The dedicated experiment flight programme. How to run it, what it measures, how to use what it finds.
Part VIII — Advanced Systems	The active damping, descent rate controller, precision scoring, and water operations software in technical detail.
Part IX — Reference	Command tables, hypothesis library, pin assignments, file listing.

SECTION 3 — SAFETY PHILOSOPHY

3.1 The Three Rules

Every decision in the MCO-1D design follows three rules in priority order:

Rule 1: The aircraft must be recoverable	No automatic action, no optimizer decision, and no experimental configuration may put the aircraft into a state it cannot recover from. The safety gates in the software are hard limits that cannot be overridden from the ground station or by the optimizer.
Rule 2: The pilot decides	The warning systems inform and the optimizer suggests, but the pilot makes every consequential decision. The margin tracker warns but does not act. The wing simulation requests approval before every test. The optimizer promotes discovered configurations but the pilot approves them before use. There is no autonomous mode that removes the pilot from the loop.
Rule 3: One new thing at a time	The commissioning sequence is designed so that each test builds on demonstrated results from the previous test. First flight before forward flight. Forward flight before discovery mode. Discovery mode before water operations. Never add two unknowns at the same time.

3.2 What the Software Will Not Do

■ WARNING: The optimizer will never command a frequency above 3.0 Hz. It will never command an amplitude above 100 degrees. It will never command an AoA that the stall protection system has determined is unsafe at the current airspeed. These are hard coded limits that cannot be changed at runtime.

■ WARNING: The wing simulation system will never inject a real wing failure into a flying aircraft. Simulated failures are only applied when the pilot has explicitly approved the test, the aircraft is above 50 metres altitude, and conditions meet all safety gates.

✓ NOTE: The active damping and precision descent systems have gain constants marked CALIBRATE AT TC-02. These systems work from estimated values until TC-02 provides measured aircraft response data. The estimates are conservative — they will work, but not optimally, until calibrated.

PART II — HARDWARE

SECTION 4 — WING AND AIRFRAME DESCRIPTION

4.1 Wing Numbering

Wings are numbered W1 through W4 from front to rear. Each number refers to a pair: W1 is the frontmost pair, W4 is the rearmost. Within each pair, port is the left wing (when facing forward) and starboard is the right wing. W1 port and W1 starboard together make up wing pair 1. This notation is used throughout the software and all log files.

W1 — forward pair	Leads the metachronal cascade. Highest aerodynamic load in forward flight.
W2 — second pair	Follows W1 with a phase offset. Primary lift generator in cruise.
W3 — third pair	Follows W2. Captures wake energy from W2.
W4 — rear pair	Closes the cascade. Efficiency and trim role. Analogous to the seal's small hind flippers — optimised for endurance rather than peak thrust.

4.2 Wing Profile

Each wing uses the S1223 high-lift aerofoil section with a chord of 165 mm. The S1223 is optimised for low Reynolds number operation and delivers high lift coefficients at the moderate angles of attack used during flapping. At the MCO-1D operating Reynolds numbers of 28,000 in hover to 89,000 in cruise, the S1223 reaches maximum lift coefficient at approximately 18 degrees angle of attack and maintains attached flow up to approximately 22 degrees. Above 22 degrees the flow separates and lift drops sharply. The stall protection system uses these values.

4.3 Sector Wheel and Stroke Limits

Each wing is driven through a sector wheel — a partial gear that converts motor rotation into a flapping arc. The sector wheel has a mechanical limit of 100 degrees total arc. The software limits flapping amplitude to 96 degrees to maintain a 4-degree safety margin from the mechanical stop. The sector can be repositioned to shift where in the stroke the wing spends most of its time, which is how the steep approach and airbrake modes work.

SECTION 5 — MOTOR AND DRIVE SYSTEM

Motor model	GoolRC 5010-14, also designated GoolRC 5010 360KV
Motor type	Brushless outrunner, 12N14P configuration
KV rating	360 KV (rpm per volt, no load)
Pole pairs	7 (required for ODrive configuration)
Gear reduction	7:1 rope drive from motor to sector wheel
Instances	4 motors total, one per wing pair
Mounting	One motor per pair drives both port and stbd wings on that pair through a shared rope drive
ODrive velocity limit	150 rad/s (to be set and verified at TC-01)

5.1 The Rope Drive

Each motor drives its wing pair through a 7:1 rope drive — a length of cord or rope wrapped around a small drum on the motor shaft and a larger drum on the sector wheel axle. The 7:1 ratio means the sector wheel turns once for every seven motor rotations. This ratio trades speed for torque, allowing the motor to overcome the aerodynamic loads on the wings without running at excessive current. The rope drive also acts as a mechanical fuse: if a wing jams, the rope slips rather than destroying the motor or the airframe.

● CAUTION: The rope drive requires periodic inspection. A fraying rope will show as a gradual increase in motor current for the same commanded frequency. The soft fault detection system will flag this as a degradation signature before it becomes a failure.

SECTION 6 — SERVO SYSTEM

6.1 Pitch Servos (Angle of Attack Control)

Each wing has one pitch servo that rotates the wing about its spanwise axis, changing its angle of attack. The angle of attack determines how much lift and drag the wing generates at a given speed and amplitude. The pitch servo is the mechanism behind most of the biomimetic hypotheses — controlling angle of attack independently on each wing is something no conventional aircraft can do.

Servo model	TORQ TCM-1001 or equivalent high-speed digital
Supply voltage	7.4 V (2S LiPo direct to servo rail)
PWM frequency	333 Hz (high-speed mode required)
Instances	8 pitch servos, one per wing
GPIO pins	W1p: 18, W1s: 23, W2p: 24, W2s: 25, W3p: 12, W3s: 16, W4p: 20, W4s: 21
Resolution	0.1 degrees — sufficient for all AoA control needs

6.2 Sweep Servos (Fore-Aft Position Control)

Each wing has one sweep servo that moves the wing fore and aft along the aircraft axis. In normal flight the sweep servo sets a fixed position. In some biomimetic modes (particularly the sea turtle figure-eight stroke, H-16) the sweep servo moves dynamically during each stroke, creating a figure-eight wing path that converts some lift into thrust.

Servo model	DS3225 or equivalent high-torque digital
Supply voltage	7.4 V
PWM frequency	50 Hz (standard)
Instances	8 sweep servos, one per wing
GPIO pins	W1p: 4, W1s: 17, W2p: 27, W2s: 22, W3p: 5, W3s: 6, W4p: 13, W4s: 19

SECTION 7 — ODRIVE CONTROLLERS

The MCO-1D uses four ODrive motor controllers, one per wing pair, connected in a CAN-FD bus. CAN-FD is a communication standard that allows multiple devices to share a single pair of wires. The Raspberry Pi sends commands to all four ODrives simultaneously over this bus, and each ODrive controls the motor for its wing pair.

Quantity	4 ODrive controllers
Node IDs	0, 1, 2, 3 (W1 through W4 respectively)
Bus type	CAN-FD
Bus speed	1 Mbit/s
Pi interface	MCP2515 SPI-to-CAN adapter on Pi SPI0
Velocity limit	150 rad/s — must be set via mco_odrive_setup.py
Control mode	Velocity control with current limiting
Startup script	mco_odrive_setup.py — run once after assembly

■ WARNING: The ODrive velocity limit must be set to 150 rad/s before any powered testing. The factory default is lower and will prevent reaching the target operating frequency. Run mco_odrive_setup.py and verify with mco_can_setup.sh before TC-01.

SECTION 8 — RASPBERRY PI AND SENSORS

8.1 The Flight Computer

The Raspberry Pi 4 is the main flight computer. It runs the Wave Library at 50 Hz — 50 times per second — reading all sensors, computing the optimal wing configuration, and sending commands to all eight servos and four ODrives. The 4 GB RAM version is recommended because the Wave Library and its data registries occupy several hundred megabytes when fully loaded after extended discovery flights.

Model	Raspberry Pi 4 Model B, 4 GB RAM recommended
OS	Raspberry Pi OS Lite (64-bit) — no desktop environment
Storage	Class 10 microSD, 32 GB minimum, 64 GB recommended
Control loop	50 Hz (20 ms per tick)
Boot time	Approximately 15 seconds to ready state
Service name	mco-flight.service (auto-starts on boot)

8.2 Sensor Suite

IMU	MPU-6050 or equivalent via I2C. Provides roll, pitch, yaw rates and accelerations at 100 Hz. Primary attitude reference.
Barometer	BMP280 or equivalent via I2C. Provides barometric altitude above launch point. Used for height reference above approximately 3 m.
Rangefinder	TF-Luna or equivalent lidar. Provides height above ground below 10 m at 50 Hz. Primary height reference for landing and approach. Over water: reads water surface height.
OAT sensor	DS18B20 waterproof temperature sensor on GPIO4 (One-Wire bus). Measures outside air temperature for density altitude computation. Approximately 3 USD. Required for high-altitude SAR.
Load cell	HX711 amplifier with strain gauge load cell. DT pin: GPIO 26, SCK pin: GPIO 7. Used for TC-02 static thrust measurement only.
GPS	Connected to Pixhawk TELEM2. Provides position data for the discovery mode racetrack pattern.

8.3 Installing the OAT Sensor

To enable the DS18B20 temperature sensor: connect the sensor data wire to GPIO4, power to 3.3V, ground to GND, with a 4.7kΩ pull-up resistor between data and 3.3V. Then add the following line to /boot/config.txt:

```
dtoverlay=w1-gpio,gpiopin=4
```

After reboot, verify with:

```
cat /sys/bus/w1/devices/28-*/w1_slave
```

The temperature reading will appear at the end of the output. If no device is found, check the wiring and the pull-up resistor.

SECTION 9 — RC RECEIVER AND PIXHAWK

The pilot flies the aircraft through a standard RC transmitter and receiver. The Pixhawk flight controller acts as a sensor bridge — it does not run the flight control algorithm, but it collects GPS, compass, and additional IMU data and sends it to the Raspberry Pi.

RC receiver	SBUS or IBUS protocol to Pi GPIO
Channel 1	Roll stick
Channel 2	Pitch stick
Channel 3	Lever (primary power/brake control)
Channel 4	Yaw stick
Pixhawk	Connected to Pi UART (TELEM2 at 921600 baud)
MAVLink bridge	mco_mavlink.py — runs as separate process
Verify channels	Run mco_odrive_setup.py and move sticks to confirm channel mapping before first flight

● CAUTION: Verify the RC channel mapping before TC-02. An incorrectly mapped channel will cause loss of control. The expected mapping is CH1=roll, CH2=pitch, CH3=lever, CH4=yaw. If your transmitter uses a different mapping, update PITCH_PINS in the library.

SECTION 10 — POWER SYSTEM

Main battery	4S LiPo, minimum 4000 mAh recommended. Provides power to ODrives and motors.
Servo battery	2S LiPo, 1000-2000 mAh. Dedicated supply for pitch and sweep servos at 7.4V nominal.
Pi power	5V 3A BEC from main battery or separate 2S supply. The Pi must have a clean 5V supply — a shared BEC with motor noise can cause SD card corruption.
Low battery exit	Discovery mode and wing simulation exit automatically at 20% main battery remaining.
Battery monitoring	Bus voltage read from ODrive telemetry. Displayed on panel and ground station.
Fusing	Blade fuse on main battery lead recommended. 30A for the main bus, 5A for Pi supply.

SECTION 11 — WIRING AND CAN-FD BUS

11.1 CAN-FD Bus Wiring

The four ODrive controllers connect in a chain on the CAN-FD bus. Each ODrive has two CAN connectors — one in, one out — allowing daisy-chaining. The chain must be terminated at both ends with a 120-ohm resistor between CAN-H and CAN-L.

```
Pi SPI0 → MCP2515 → ODrive 0 → ODrive 1 → ODrive 2 → ODrive 3 → 120Ω terminator
```

11.2 Servo Wiring

Pitch servos connect directly to the Pi GPIO pins listed in Section 6. The Pi generates PWM signals at 333 Hz for pitch and 50 Hz for sweep. All servo grounds must connect to the Pi ground as well as the servo battery ground. Do not connect servo power to the Pi 5V rail — the current draw would damage the Pi.

11.3 Running the CAN Setup Script

After wiring, configure the Pi CAN interface by running:

```
sudo bash /home/pi/mco_can_setup.sh
```

This script brings up the can0 interface at 1 Mbit/s and sets it to start automatically on boot. Verify it is working with:

```
ip link show can0
```

The output should show the interface as UP with a state of UNKNOWN (this is normal for CAN interfaces).

SECTION 12 — ASSEMBLY SEQUENCE

Assemble in this order. Do not power anything until Step 10.

1	Mount airframe	Attach all structural members. Verify rigidity.
2	Mount sector wheels	One per wing pair. Verify smooth rotation.
3	Mount motors	One per wing pair. Verify shaft alignment with rope drum.
4	Install rope drives	Thread and tension. Verify 7:1 ratio by counting turns.
5	Mount wings	Attach to sector wheel pivot points. Verify free movement.
6	Install pitch servos	One per wing. Set to neutral before attaching horn.
7	Install sweep servos	One per wing. Set to neutral before attaching horn.
8	Mount ODrives	Secure to airframe. Connect motor phases. Check polarity.
9	Wire CAN bus	Daisy chain ODrives 0→1→2→3. Install terminators.
10	Mount Pi and sensors	Secure Pi. Connect IMU, baro, rangefinder via I2C.
11	Connect servo harness	Pitch servos to GPIO pitch pins. Sweep to sweep pins.
12	Wire power	Main battery to ODrives. Servo battery to servo rail. BEC to Pi. Add fuses.
13	Install RC receiver	SBUS to Pi UART. Bind to transmitter.
14	Connect Pixhawk	TELEM2 UART to Pi at 921600 baud.
15	Install OAT sensor	DS18B20 to GPIO4 with 4.7kΩ pull-up.
16	First power-on	Pi only, no motors. Verify OS boots. Run mco_install.sh.
17	ODrive setup	Run mco_odrive_setup.py. Set vel_limit=150.
18	Servo calibration	Run servo calibration sequence. Verify all 16 servos move.
19	Fill measurement sheet	Record all physical dimensions in Build_Measurement_Sheet.
20	TC-01 bench test	See Section 26.

PART III — SOFTWARE INSTALLATION

SECTION 13 — RASPBERRY PI OPERATING SYSTEM SETUP

13.1 Installing the OS

Download the Raspberry Pi Imager from raspberrypi.com. Select Raspberry Pi OS Lite (64-bit) — the version without a desktop environment. This keeps the system lean and ensures the control loop has maximum CPU time. Write to a 32 GB or larger microSD card.

Before writing, open the imager's advanced settings and:

Set hostname	mco-1d (or your preferred name)
Enable SSH	Required for ground station connection
Set username	pi (the software expects this)
Set password	Choose a strong password
Set WiFi	Your field network SSID and password
Set locale	Your timezone and keyboard layout

13.2 First Boot Configuration

Insert the SD card, connect to a monitor and keyboard for first boot. After login, run:

```
sudo raspi-config
```

In the configuration tool: enable I2C (for IMU and barometer), enable SPI (for the MCP2515 CAN adapter), enable the serial port but disable the serial console (for Pixhawk), and expand the filesystem. Reboot.

13.3 Adding the CAN Overlay

Add this line to /boot/config.txt to enable the MCP2515 CAN adapter:

```
dtoverlay=mcp2515-can0,oscillator=8000000,interrupt=25
```

And for the OAT sensor:

```
dtoverlay=w1-gpio,gpiopin=4
```

Reboot after editing config.txt.

SECTION 14 — INSTALLING THE WAVE LIBRARY

14.1 One-Step Installation

The mco_install.sh script installs all Python dependencies and copies all software files to the correct locations. Run it once after first boot:

```
bash mco_install.sh
```

This installs: pigpio (servo PWM), python-can (CAN bus), and creates the directory structure at /home/pi/knowledge/ for persistent data storage. It also installs the systemd service files so the software starts automatically on boot.

14.2 Software Files

mco_wave_library.py	Main flight control library — 42,115 lines. This is the brain of the aircraft.
mco_main.py	Flight loop entry point. Runs at 50 Hz.
mco_mavlink.py	Pixhawk sensor bridge. Runs as separate process.
mco_gs_server.py	Ground station TCP server. Port 5760.
mco_logger.py	Flight data logger. 50 Hz CSV output.
mco_odrive_setup.py	ODrive configuration tool. Run once after wiring.
mco_panel.py	Touchscreen panel driver. 4-button + status bar.
mco_ground_station.py	Laptop ground station client.
mco_sync.py	Knowledge transfer between Raspberry Pis.
mco_hypothesis_tool.py	Tool for injecting new biological hypotheses.
mco_loadcell.py	HX711 load cell reader for TC-02 (to be written).
mco_can_setup.sh	CAN interface configuration shell script.
mco_install.sh	One-step installation script.
mco-flight.service	Systemd service file for auto-start.
mco-panel.service	Systemd service file for panel.

SECTION 15 — ODRIVE CONFIGURATION

After the CAN bus is wired and the software is installed, configure the ODrives by running:

```
python3 mco_odrive_setup.py
```

This script connects to each ODrive over CAN, sets the following parameters, and saves them to non-volatile memory:

Motor type	BLDC (brushless DC)
Pole pairs	7 (GoolRC 5010-14 confirmed)
Current limit	Set from Build Measurement Sheet
Velocity limit	150 rad/s — critical for frequency ceiling
Control mode	Velocity control
Encoder type	Hall effect (if fitted) or sensorless

■ WARNING: vel_limit=150 rad/s must be set before any powered wing test. The default factory setting is lower and will prevent the wings from reaching the operating frequency range. Verify the setting was saved by power-cycling the ODrive and re-reading the parameter.

SECTION 16 — GROUND STATION SETUP

The ground station is a laptop running mco_ground_station.py that connects to the aircraft over WiFi. The aircraft runs a TCP server on port 5760. The ground station sends commands and receives real-time telemetry.

```
python3 mco_ground_station.py --host mco-1d.local --port 5760
```

The ground station displays: current flight mode, wing configuration, all six margin indicators, discovery mode status, fault summary, and the live optimizer score. Commands are typed in the command box or selected from menus.

16.1 Verifying the Connection

With the aircraft powered and the software running, connect the ground station and send:

```
arm
```

The aircraft should respond with a status message confirming arm state. If no response within 5 seconds, check WiFi connection and verify the mco-flight.service is running on the Pi:

```
sudo systemctl status mco-flight.service
```

SECTION 17 — VERIFYING THE INSTALLATION

Run these checks in order before proceeding to TC-01.

V-01	Wave Library imports	python3 -c "import mco_wave_library; print('OK')" — must print OK
V-02	CAN bus active	ip link show can0 — must show UP
V-03	ODrive responds	python3 mco_odrive_setup.py --status — must show all 4 ODrives
V-04	IMU readable	python3 -c "import mco_mavlink; print(mco_mavlink.read_imu())"
V-05	Servos move	python3 mco_odrive_setup.py --servo-test — all 16 servos deflect
V-06	OAT sensor	cat /sys/bus/w1/devices/28-*/w1_slave — shows temperature
V-07	Ground station connects	Run mco_ground_station.py — status messages appear within 5 seconds
V-08	Log file created	After 10 seconds running, check /home/pi/logs/ for CSV file

PART IV — THE FLIGHT CONTROL SYSTEM

SECTION 18 — ARCHITECTURE OVERVIEW

18.1 The 50 Hz Control Loop

Every 20 milliseconds the flight computer executes one tick of the control loop. In each tick it reads all sensors, computes the best wing configuration for the current conditions, sends commands to all eight servos and four ODrives, logs the data, and updates all monitoring systems. Fifty ticks per second gives the aircraft adequate bandwidth to respond to gusts and maintain stable flight.

The tick is structured as a chain of 53 software layers, each adding one capability. The layers run in a fixed order: safety checks first, then core flight control, then optimization, then advanced features. Any layer can modify the wing commands produced by the layers before it, but safety layers run last and cannot be overridden.

18.2 The Six Flight Phases

Takeoff/Sprint	Maximum performance. Lever above 88%. Only raw thrust score matters. Full amplitude.
Transit	Efficient forward flight. Lever 60-88%. Range per watt-hour weighted heavily.
Cruise	Moderate forward flight. Lever 50-60%. Balanced efficiency and precision.
SAR Loiter	Slow hover-like flight over a target. Endurance and stability weighted equally.
Approach	Descent to landing. Lever 12-50%. Precision weighted most heavily.
Hover	Near-stationary flight. Lever near 50%. Minimum power and maximum stability.

SECTION 19 — THE LEVER — PRIMARY FLIGHT CONTROL

The lever is Channel 3 on the RC transmitter — the throttle stick on a conventional helicopter. On the MCO-1D it works differently from any conventional aircraft. The lever maps to a complete wing configuration, not just a power level.

Lever 100% (full forward)	Maximum thrust. Full amplitude, maximum frequency, optimal cascade. Sprint and takeoff mode.
Lever 88-100%	Performance zone. Only thrust score matters. The demanding input rule applies.
Lever 50-88%	Normal flight zone. Phase-indexed scoring applies. Optimizer balances performance with mission objective.
Lever 50%	Neutral in flight — hover or trim.
Lever 12-50%	Approach and airbrake zone. Precision scoring weighted highest.
Lever 0-13%	Steep approach zone. Pigeon-wing dihedral activates automatically. Glide slope ranges from 5° (lever 13%) to 30° (lever 0%).
Lever 0% (full aft)	Maximum aerodynamic braking. Wings at maximum dihedral, sector shifted to generate drag.

■ WARNING: Forward is always power. This convention is absolute and never changes regardless of mode. Lever forward = more thrust. Lever aft = less thrust or active braking.

19.1 The Demanding Input Rule

When the pilot makes a demand — any stick above 70% deflection, lever above 88% or below 12%, or any abrupt control movement — the optimizer immediately switches to maximum performance mode. Precision scoring, endurance weighting, and efficiency calculations are all suspended. The aircraft gives everything it has. When the pilot returns to steady inputs, the optimizer resumes its normal multi-objective scoring after a 3-second settling period.

SECTION 20 — WING CONFIGURATION AND THE CASCADE

20.1 The Metachronal Cascade

The four wing pairs do not all flap at the same time. W1 leads, then W2 follows with a phase offset, then W3, then W4. This staggered timing — called a metachronal cascade — means each rearward pair can capture energy from the wake shed by the pair ahead. Plesiosaur research (Muscutt et al. 2017) demonstrated that hind flippers generated 60% more thrust and 40% better efficiency when correctly phased with the forward flippers compared to operating alone. The cascade phase is the angle offset between consecutive pairs.

20.2 What the Optimizer Controls Per Wing

Flapping amplitude	How far the wing moves in its arc. Maximum 96 degrees (4° from mechanical limit).
Flapping frequency	How many strokes per second. Range 0.5-3.0 Hz. Hover estimated near 2.5 Hz.
Angle of attack	The pitch of the wing surface relative to airflow. S1223 CL_max at 18°. Post-stall above 22°. Eight wings with independent AoA control — this capability has no equivalent in any conventional aircraft.
Sweep angle	Fore-aft position of the wing. Fixed in normal flight, oscillating in H-16 figure-eight stroke mode.
Cascade phase	Timing offset relative to the lead pair. Speed-adaptive — changes with airspeed per plesiosaur research.

SECTION 21 — THE OPTIMIZER — HOW THE AIRCRAFT LEARNS

21.1 What the Optimizer Does

The optimizer is the system that learns what wing configurations work best. Every flight, it continuously tries variations on the current best-known configuration, measures the result, and either promotes the variation if it is better or discards it if it is worse. Over many flights the optimizer builds a map of the best configuration for every lever position and every flight condition. This map is called the discovery registry, and it persists between flights on the SD card.

21.2 The Four-Layer Exploration System

Layer 1 — Gradient	Always running. Makes small improvements to the current best configuration. Like tuning a radio: turn the dial until it gets better, then hold there.
Layer 2 — Biomimetic	Tests biological hypotheses from the 23-hypothesis library. Each hypothesis is a configuration inspired by a specific animal's locomotion — bat, plesiosaur, sea turtle, etc.
Layer 3 — Evolutionary	Breeds new configurations by combining elements of two previously discovered good configurations, like crossing two well-performing plant varieties.
Layer 4 — Random	Explores the full parameter space randomly. Finds configurations that gradient search would never reach because they require moving through a worse region to get to a better one.

21.3 The Three-Tier Objective System

Tier 1 — Safety envelope	Hard limits. The optimizer cannot generate any configuration that violates these. Frequency cap, amplitude cap, stall protection.
Tier 2 — Phase objectives	The mission goal for the current flight phase. Transit: maximise range per watt-hour. SAR loiter: maximise endurance and stability. Approach: maximise precision. Takeoff: maximise thrust.
Tier 3 — Envelope expansion	5% of exploration time pushes beyond the current demonstrated envelope in small steps. This is how new capabilities are discovered — the aircraft finds out it can do more than it currently does.

SECTION 22 — BIOMIMETIC HYPOTHESIS LIBRARY

The library contains 23 active hypotheses drawn from 11 biological sources and 93 total configuration variants. Each hypothesis is a specific configuration inspired by the locomotion of a real animal, translated from water or other-scale flight into parameters appropriate for the MCO-1D operating conditions.

22.1 Why Biological Research Matters

Animals have had millions of years to optimise their locomotion by natural selection. A dragonfly's tandem wings, a plesiosaur's four flippers, a sea turtle's figure-eight stroke, and a bat's speed-adaptive upstroke all represent solutions to the same fundamental problem the MCO-1D faces: how to move efficiently through a fluid using flapping surfaces. The optimizer cannot discover these solutions by gradient search alone because they require moving through worse configurations to reach them. The hypothesis library gives the optimizer a direct path.

22.2 Reynolds Number and Medium Corrections

All biological data has been corrected for the MCO-1D operating conditions before entry into the library. Two types of correction apply:

Medium correction	Aquatic animals operate in water (density 1025 kg/m³). The MCO-1D operates in air (density 1.225 kg/m³ at sea level). Absolute force values do not transfer. Dimensionless parameters (phase ratios, timing fractions, Strouhal number) transfer directly.
Reynolds number correction	The MCO-1D operates at Re 28,000 (hover) to 89,000 (cruise). Large pterosaurs operated at Re 1,000,000+. Their optimal AoA values are not valid at MCO-1D Re. The S1223 polar at Re 50,000 gives CL_max at 18°, not at 32° as the pterosaur data suggested. All AoA values have been corrected to the S1223 polar.

22.3 The Three Novel Hypotheses — New Territory

H-24, H-25, and H-26 are MCO-1D originals with no direct biological precedent. They arise from the unique capability of having independent AoA control on each of eight wings — something no conventional aircraft possesses:

H-24 — Spanwise AoA gradient	W1 at force-vectoring AoA (15°, maximum thrust) while W4 at high-lift AoA (22°, maximum vertical lift). Front pairs drive forward while rear pairs support weight. Breaks the thrust-lift coupling that limits all fixed-pitch flapping wings.
H-25 — Per-stroke AoA roll	AoA differential between port and starboard during the downstroke creates roll authority. The pitch servo responds in under 50ms. This is approximately 4× faster than amplitude-based roll through the motor and rope drive.
H-26 — Force vector decoupling	Port wings at higher AoA than starboard creates lateral force without banking. If the active damping system holds the fuselage level, the aircraft may translate sideways while remaining level — hovering sideways next to a cliff face without attitude change. Entirely unknown until TC-02 flight testing.

SECTION 23 — FAULT TOLERANCE SYSTEM

23.1 Design Philosophy

The fault tolerance system is based on the aviation principle that more fatal accidents come from attempting maximum performance with a degraded aircraft than from accepting a controlled landing. When a wing fails, the system aims for controlled flight to a safe landing, not maximum remaining thrust.

23.2 What Happens When a Wing Fails

Detection	Ten consecutive ticks of zero motor current confirms a failure. The 0.2-second confirmation window prevents false positives from brief electrical interruptions.
Immediate response	The failed wing is stowed: amplitude zero, swept aft to minimum drag, angle of attack neutral, sector locked. The motor stops.
Partner wing	The partner wing on the opposite side of the same pair KEEPS FLYING. Port and starboard have no aerodynamic interaction. Stopping the partner wastes 12.5% of remaining lift capacity for no benefit.
Compensation	Remaining healthy wings increase frequency modestly. AoA trim adjusts to correct roll and pitch moments. Cascade resequences if a complete pair is lost.
Pilot notification	Aviation-standard urgency language: CONTINUE WITH CAUTION (single wing, 7 remaining), LAND AS SOON AS PRACTICABLE (two wings, manageable), LAND AS SOON AS POSSIBLE (four wings, thin margins), LAND IMMEDIATELY (uncontrollable configuration).

23.3 Pre-Flight Failure Briefing

Before any flight in challenging conditions, the pilot can query the 255-combination failure table:

```
failure_briefing
```

This shows: for each number of failed wings, the worst-case stall speed increase, service ceiling reduction, and urgency level. All 255 combinations were pre-computed at library startup.

SECTION 24 — FLIGHT MARGIN TRACKER

The U-2 spyplane at 70,000 feet had only five knots between stall speed and never-exceed speed. The margin tracker was designed with this coffin corner lesson in mind: warn the pilot specifically and early, never act autonomously.

Speed margin	Distance from current airspeed to demonstrated Vmax.
Stall margin	Distance above stall speed, corrected for bank angle. In a 45° bank, stall speed is 1.19× the level value.
Frequency margin	Distance from current frequency to 3.0 Hz ceiling.
Amplitude margin	Distance from current amplitude to 96° limit.
Gust margin	Current turbulence vs worst conditions survived, combined with amplitude headroom for gust response.
Combined margin	The minimum of all five. The coffin corner metric.

■ WARNING: The margin tracker warns only. It does not take automatic action. The only autonomous effect is that the exploration system pauses when margins are narrow. All control remains with the pilot.

SECTION 25 — GROUND OPERATIONS STATE MACHINE

The aircraft has a formal ground state machine that prevents unsafe sequences. States progress in order — you cannot skip to flight without passing through pre-flight checks.

PARKED	Power on, all wings stowed. Initial state.
PRE-FLIGHT	Checklist running. Systems verified. Initiated by pressing PRE-FLIGHT button.
ARMED	Ready for flight. Wings at neutral. Motors ready to respond to lever input.
FLIGHT	Airborne. Full flight control active.
COOLDOWN	8-minute post-flight motor cooling period.
POST-FLIGHT	Data export and system shutdown sequence.

✓ NOTE: The 4-button touchscreen panel shows current state and allows state transitions. The panel also shows the six margin bars (yellow/amber/red) as a continuous warning system during flight.

PART V — COMMISSIONING TESTS

SECTION 26 — TC-01: BENCH TEST — HARDWARE VERIFICATION

TC-01 is a ground test. The aircraft does not leave the bench. No altitude, no flight, no movement through air. The purpose is to verify that every piece of hardware and software works correctly before applying any aerodynamic loads. All subsequent tests depend on TC-01 passing completely.

26.1 Prerequisites

Before beginning TC-01, the following must be complete: all V-01 through V-08 verification checks passed, Build_Measurement_Sheet filled with all physical dimensions, ODrive vel_limit confirmed at 150 rad/s, RC channel mapping verified, all 16 servos moving in the correct direction, ground station connected and receiving telemetry.

26.2 TC-01 Test Sequence

TC-01-01	Servo neutral verification	All 16 servos at commanded neutral. Measure actual angle. Record deviation. Adjust trim offsets in constants if >2°.
TC-01-02	Servo range verification	Command each servo to full positive and full negative. Verify travel without binding. Record mechanical limits.
TC-01-03	ODrive communication	Verify all 4 ODrives respond to CAN queries. Log node IDs and firmware versions.
TC-01-04	Motor spin test — low power	Command each ODrive to 10 rad/s. Verify motors spin. Listen for bearing noise. Check for unusual vibration.
TC-01-05	Wing flapping at low frequency	Command 0.5 Hz on all pairs. Verify smooth mechanical motion. Check rope drives for slipping or fraying. Verify sector wheels reach full arc without binding.
TC-01-06	Wing flapping at operating frequency	Command 2.0 Hz on all pairs. Run for 60 seconds. Check motor temperatures. Check rope drive tension. Verify all servos tracking correctly at frequency.

TC-01-07	Frequency ceiling bench test	Step frequency from 2.0 to 3.0 Hz in 0.1 Hz steps. Hold 30 seconds at each step. Record motor current and temperature. Identify the maximum sustained frequency before thermal limiting. Record as the operational frequency ceiling.
TC-01-08	RC input test	Move all sticks to extremes. Verify correct mapping: CH1=roll, CH2=pitch, CH3=lever, CH4=yaw. Verify centre positions map to 50% lever and zero stick.
TC-01-09	Safety system test	Command frequency above 3.0 Hz — verify hard cap activates. Command amplitude above 96° — verify hard cap activates. Verify fault detection fires when a motor lead is unplugged.
TC-01-10	Load cell static thrust	Secure aircraft to load cell fixture. Command hover estimated frequency (2.5 Hz). Record thrust. Vary frequency 1.5-3.0 Hz, record thrust at each. This produces the lift-frequency curve needed for TC-02 calibration.
TC-01-11	Fill measurement sheet	Record all measured values: actual hover thrust, frequency ceiling, motor temperatures at ceiling, rope drive condition, servo trim offsets. Update constants in wave library from measured values.

26.3 TC-01 Pass Criteria

TC-01 is passed when all of the following are true:

All servos	Move to commanded position within ±2°
All ODrives	Respond to CAN commands, no error codes
All motors	Spin without unusual noise or vibration
Frequency ceiling	Aircraft can sustain at least 2.5 Hz for 60 seconds
Motor temperature	Below 60°C at operating frequency after 60 seconds
Static thrust	Greater than MTOW (33.8 N) at 2.5 Hz
Safety systems	All hard caps confirmed active
RC mapping	All channels verified correct
Measurement sheet	Fully populated with measured values

■ WARNING: Do not proceed to TC-02 unless all TC-01 pass criteria are met. If static thrust is less than MTOW at the frequency ceiling, the aircraft cannot hover. Identify and resolve the discrepancy before any flight attempt.

SECTION 27 — TC-02: HOVER TEST — FIRST CONTROLLED FLIGHT

TC-02 is the first flight. The goal is not impressive performance — the goal is controlled, stable hover for a measured time at a measured altitude. Every advanced capability in the Wave Library depends on data that can only be obtained from TC-02: actual hover frequency, lift-frequency curve in air, roll response to differential amplitude, and angular momentum characteristics. Take TC-02 seriously.

27.1 Site Requirements

Open area	Minimum 50m × 50m clear of obstacles. No people within 30m during flight.
Wind	Calm or very light wind only. Below 5 knots. Do not attempt TC-02 in gusty conditions.
Altitude	Sea level to 300m elevation. Do not attempt TC-02 at high altitude — the density altitude compensation is not yet calibrated.
Recovery space	Soft ground or grass preferred. The aircraft will likely make some uncontrolled descents during initial hover tuning.
Ground station	Operator at laptop ground station, not holding transmitter. Separate pilot on transmitter. Two people minimum.

27.2 TC-02 Pre-Flight

Run the normal pre-flight procedure (Section 29). Additionally: verify the calibrate_descent_controller() function is ready to call with measured data, verify the active damping calibration function is ready, attach the load cell fixture if a second hover measurement is planned.

27.3 TC-02 Test Sequence

TC-02-01	First hover attempt	Increase lever slowly from 50% toward 70%. Hold at each position for 3 seconds. Record the lever position and frequency when lift-off occurs. This is the hover lever position. Land. Repeat 3 times.
TC-02-02	Hover stability assessment	Achieve hover at measured lever position. Hold for 30 seconds without correcting. Record altitude deviation (sigma_h). If sigma_h exceeds 0.5m, investigate before continuing.
TC-02-03	Controlled altitude hold	Hover for 60 seconds with ground station monitoring altitude. Pilot makes gentle corrections. Record hover frequency from telemetry. This is the key TC-02 measurement: actual hover frequency.

TC-02-04	Lift-frequency curve in air	At constant altitude, vary frequency ±0.3 Hz from hover frequency. Record altitude change rate at each frequency step. This gives dL/df — the lift-frequency slope needed for descent controller calibration.
TC-02-05	Roll response test	From stable hover, apply small roll stick input (20%). Record roll rate achieved. Release stick. Record time to settle. Repeat at 40% and 60% stick. This calibrates the active damping gains.
TC-02-06	Calibrate descent controller	With measured hover frequency and lift-frequency slope, run: calibrate_descent_controller(hover_freq_hz=X, lift_freq_slope_N_per_Hz=Y). Verify 100 fpm descent is achievable and stable.
TC-02-07	Calibrate active damping	Use measured roll response data to set ACTIVE_DAMP_K_AMP and ACTIVE_DAMP_SETTLE_TIME_S to measured values.
TC-02-08	First optimizer discovery	Enable auto-discovery. Hover for 5 minutes. Allow the optimizer to begin building its hover configuration map. Export knowledge at end of session.

27.4 TC-02 Critical Outputs

The following measured values from TC-02 enable all subsequent systems. Record them carefully and update the Wave Library constants.

Hover frequency	The frequency at which the aircraft maintains altitude at the measured MTOW. Estimated 2.5 Hz. Could be 1.8-3.2 Hz.
Lift-frequency slope	How much lift changes per Hz of frequency change near hover. In N/Hz. Used by descent controller.
Roll rate	Degrees per second of roll produced by 20%, 40%, 60% differential amplitude command.
Settle time	Time from roll input removal to roll rate below 2°/s. Used by active damping calibration.
Angular momentum	Whether the aircraft exhibits significant gyroscopic precession during attitude changes.
Sidestep observation	During H-26 test: does differential AoA produce lateral translation or roll? This is the most uncertain prediction.

SECTION 28 — TC-03: TRANSITION AND FORWARD FLIGHT

TC-03 begins only after TC-02 is complete and all calibrations are done. The goal is to demonstrate controlled transition from hover to forward flight and back, and to measure performance across the full speed range.

28.1 TC-03 Prerequisites

TC-02 complete and all calibrations applied. Descent controller verified at 100 fpm. Active damping calibrated and tested. Discovery registry contains at minimum 20 promoted configurations. Pilot comfortable with hover control from TC-02.

28.2 TC-03 Test Sequence

TC-03-01	Hover-to-forward transition	From stable hover, gradually advance lever toward 70%. Allow pitch forward to develop naturally. Record transition speed and lever position. Return to hover. Repeat 3 times.
TC-03-02	Forward flight at constant altitude	Achieve forward flight at estimated cruise lever (65%). Hold altitude using descent controller. Record actual airspeed from Pixhawk GPS. Record power consumption.
TC-03-03	Speed envelope sweep	Fly at 60%, 65%, 70%, 75%, 80% lever. Record airspeed and power at each. Identify stall speed in forward flight from minimum stable speed. Record VNE if approached.
TC-03-04	Controlled descent in forward flight	Engage descent_ctrl_engage 100 in forward flight. Verify 100 fpm descent is maintained. Verify 300 fpm descent when commanded. This validates the IFR descent capability.
TC-03-05	Discovery mode first run	Fly the racetrack pattern at 100 feet AGL. Engage disc_mode_engage for 15 minutes. Allow exploration block to run. Export knowledge. Review promoted configurations.
TC-03-06	CT-J curve measurement	Measure thrust coefficient versus advance ratio across speed range. This enables the stroke plane controller design planned for post-TC-03 development.

PART VI — FLIGHT OPERATIONS

SECTION 29 — NORMAL PRE-FLIGHT PROCEDURE

Perform these checks before every flight, including TC-02 and TC-03. The checklist runs automatically when PRE-FLIGHT is pressed on the panel, but the pilot must verify each item.

1	Area clear	No people within 30m. Landing zone identified.
2	Weather	Wind below 15 knots for normal ops. Below 5 knots for TC-02. Check gust forecast.
3	Battery charge	Main battery above 90%. Servo battery above 90%. Check voltage on ground station.
4	SD card	Verify /home/pi/logs/ has write space. Minimum 500 MB free.
5	ODrives armed	Ground station shows all 4 ODrives IDLE.
6	Servo check	All 16 servos respond to test commands.
7	RC signal	Ground station shows RC signal VALID.
8	IMU calibrated	Place aircraft level. Verify IMU reads 0° pitch, 0° roll.
9	Knowledge loaded	Ground station shows discovery registry status.
10	Margins nominal	All 6 margin bars green on panel.
11	Arm	Press ARM button. Verify armed state.
12	Control check	Move all sticks. Verify correct response direction.

SECTION 30 — NORMAL FLIGHT PROCEDURES

Takeoff	From armed state, advance lever smoothly from 50% toward 70%. The aircraft will lift off between 60-75% depending on conditions. Do not jerk the lever — smooth input gives the optimizer time to find the right configuration for the developing conditions.
Hover	Hold lever at the position that maintains altitude. This will be close to the hover lever position found in TC-02. Small roll and yaw inputs for positioning. The margin tracker will warn if hover frequency is approached.
Forward flight	From hover, advance lever toward 65-70%. Allow pitch to develop. The aircraft will accelerate and gain airspeed. Maintain altitude with lever position. Engage heading hold with heading_hold command if available.
Descent and landing	Reduce lever toward 50% for descent. For precision approach: descent_ctrl_engage 100 activates 100 fpm controlled descent using rangefinder. For normal landing: fly approach slope, hold off, reduce lever. Do not cut lever abruptly — the aircraft needs a few wingbeats to lose momentum cleanly.

SECTION 31 — POST-FLIGHT PROCEDURE

1	Disarm	Press DISARM. Verify motors stop.
2	Post-flight check	Press POST-FLIGHT button. 8-minute cooldown.
3	Knowledge export	Press EXPORT or run knowledge_export_usb.
4	Log retrieval	Copy /home/pi/logs/ to USB or laptop.
5	Battery check	Check main battery for puffing or heat.
6	Rope drives	Inspect all 4 rope drives for fraying or slipping.
7	Servos	Verify all 16 servos return to neutral.
8	Airframe	Visual inspection for cracks or loosening.
9	Log review	Open CSV log, verify 50 Hz sample rate throughout.

SECTION 32 — EMERGENCY PROCEDURES

32.1 Loss of Control — Immediate

■ WARNING: If the aircraft moves unexpectedly or control response is wrong: 1. Reduce lever toward 50% immediately. 2. If aircraft continues to behave unexpectedly: full lever aft. 3. Disarm if aircraft cannot be controlled. A controlled descent to hard landing is better than an uncontrolled flight that endangers people.

32.2 Wing Failure in Flight

If the fault detection system announces a wing failure: read the urgency level from the ground station. CONTINUE WITH CAUTION means land when convenient. LAND AS SOON AS POSSIBLE means land within the next 2 minutes. LAND IMMEDIATELY means put the aircraft down now, accept any surface. The fault system has already applied compensation — continue flying to the nearest safe landing point.

32.3 Loss of RC Signal

If RC signal is lost, the aircraft holds its last commanded configuration for 2 seconds then begins a controlled descent at 300 fpm. Re-establishing RC signal resumes normal control.

32.4 Battery Low Warning

At 30% main battery: amber warning on panel. Land soon. At 20% main battery: red warning. Land now. Discovery mode and wing simulation exit automatically at 20%.

SECTION 33 — BACKCOUNTRY OPERATIONS

33.1 Slope-Aware Touchdown

On a sloped surface, the rangefinder closure rate diverges from the barometric sink rate. The system detects this divergence, infers the slope angle, and corrects the height reference automatically by up to 0.5 metres. All landing systems use the corrected height.

33.2 Confined Area Turns

To execute the tightest possible turn — for canyon escape or obstacle avoidance — apply full yaw pedal and full roll stick simultaneously in the same direction, both above 80% deflection. This is the confined area minimum-radius turn. No mode switch required. Release either stick to return to normal control.

33.3 Steep Approach

The lever range 0-13% activates continuous steep approach mode. The wings adopt pigeon-wing dihedral — raising both wings above the horizontal — which creates high drag while maintaining enough lift for a controlled descent. Glide slope varies continuously: 5 degrees at lever 13%, 30 degrees at lever 0%.

SECTION 34 — WATER OPERATIONS

● CAUTION: Water operations require float hardware that is not yet specified. The software framework is complete but water operations cannot be conducted until floats are installed. Additionally, hover capability must be demonstrated at TC-02 before any water operations are planned. This section describes the planned procedure for future operations.

34.1 Why the MCO-1D Is Better Than a Conventional Seaplane

Conventional seaplanes must accelerate across the water surface to reach flying speed, fighting maximum hydrodynamic drag at the "hump" — the speed where the bow wave is at its worst. This is where porpoising occurs and where most seaplane accidents happen. The MCO-1D departs vertically, bypassing the hump entirely. Its stall speed of 4.85 m/s is lower than any fixed-wing waterborne aircraft. The glassy water altitude problem — where pilots cannot judge height above a mirror-smooth surface — is solved by the rangefinder providing an objective height reference for a 100 fpm controlled descent.

34.2 Wave Height Limits (FAA Standard)

Wave heights are measured crest to trough in inches or feet — the standard pilot measurement. Significant wave height and sea state numbers are not used.

15 inches (0.38m)	FAA safe limit for light seaplanes. Normal operations below this.
18 inches (0.46m)	FAA restriction begins. Experienced pilots only.
>20% of float length	Expert pilots only. Float dimensions required.
>50% of float height	Do not attempt. Float dimensions required.

34.3 Departure in Waves — FAA Doctrine

FAA-H-8083-23: apply power as the floats begin rising on a wave. The float rides up over the crest, already building speed as it arrives. For MCO-1D: initiate the vertical jump as the float begins to rise, so liftoff occurs at or near the crest — the highest point of the wave, requiring minimum climb to clear the water surface.

■ WARNING: Departing from the wave trough is wrong. The aircraft starts at its lowest point with an oncoming wave face directly ahead.

34.4 Glassy Water Landing

Engage descent_ctrl_engage 100. The rangefinder provides height over water. Target 100 fpm (0.508 m/s) descent rate per FAA standard. Do not judge altitude visually. Trust the rangefinder only. Continue descent until float contact. Do not flatten out.

PART VII — DISCOVERY PROGRAMME

SECTION 35 — DISCOVERY MODE: CONCEPT AND PURPOSE

35.1 Discovery Is Always Happening

The optimizer runs in every flight mode at every moment. During a transit flight, it is quietly testing small variations on the cruise configuration. During a SAR loiter, it is refining the endurance configuration. This continuous background exploration is the baseline.

Discovery Mode is what happens when the aircraft dedicates an entire flight to exploration with no competing mission demands. The aircraft holds a safe racetrack pattern at 100 feet and runs every optimizer system at maximum throughput for the duration of the flight. The optimizer gets the whole flight. This is the difference between a researcher who experiments on the side while doing other work versus one who has a dedicated laboratory day.

35.2 The Racetrack Pattern

The Discovery Mode pattern is a standard aviation holding pattern: two 30-second straight legs connected by 30-second 180-degree turns. The inbound heading is set by the pilot before engaging. One complete loop takes approximately 2 minutes.

The straight inbound leg is where experiments run — the flight is stable, the heading is known, and the aircraft is in trim. The turns are used by the envelope expansion system which specifically tests the bank angle axis. The pattern gates experiments automatically without any additional logic.

35.3 The Four Experiment Blocks

Block 0 (minutes 0-15) Pure Exploration	All four exploration layers at full throughput. Every suitable tick on the inbound leg runs a trial. At 50 Hz with a 30-second inbound leg, each leg provides 1,500 ticks. Multiple complete trial cycles run per leg.
Block 1 (minutes 15-30) Envelope Expansion	Tier 3 envelope expansion at 100% rate instead of the normal 5%. Systematic steps along the least-explored boundary axis. The aircraft finds out what it can do beyond current demonstrated limits.
Block 2 (minutes 30-45) Biomimetic Hypotheses	All 23 hypotheses and 93 variants tested systematically. A complete 60-minute flight can test all hypotheses multiple times. New hypotheses injected via mco_hypothesis_tool.py queue here.
Block 3 (minutes 45-60) Wing Simulation	Individual idle tests and progressive failure chains. Aircraft auto-climbs to 200 feet minimum before starting. Progressive chains need 250 feet. Arms next untested combination, runs trial, records result.

SECTION 36 — RUNNING A DISCOVERY FLIGHT

36.1 Engaging Discovery Mode

After completing the normal pre-flight and achieving stable flight:

```
disc_mode_engage(inbound_heading=270, duration_min=60, pattern_alt_ft=100)
```

The aircraft takes over heading control for the racetrack pattern. The pilot retains full authority over all inputs at all times. To exit: type disc_mode_exit or wait for the duration to expire. The aircraft automatically exports its knowledge when Discovery Mode exits.

36.2 What to Monitor During a Discovery Flight

Discovery Mode status	Shows current block, elapsed time, number of experiments run, number of configurations promoted.
Margin tracker	All six bars should remain green throughout. Exploration freezes automatically if margins narrow.
Battery level	Mode exits automatically at 20%. Plan to land when mode exits.
Wing sim status	Shows which failure combination is being tested, current altitude, and whether the test completed or required re-engagement.
Promoted configurations	Count increases as the optimizer finds better configurations. A productive session promotes 10-30 new configurations.

36.3 After a Discovery Flight

The knowledge export runs automatically when Discovery Mode exits. To transfer knowledge to a second Pi or to a backup:

```
knowledge_export_usb (copies to USB stick) mco_sync.py push (sends to laptop hub) mco_sync.py
transfer (Pi-to-Pi direct transfer)
```

SECTION 37 — FOCUSED DISCOVERY MODES

Instead of cycling through all four blocks, the pilot can direct the optimizer to work on a specific problem for the entire session. This is particularly valuable when weather or conditions make one type of experiment more productive than others.

gust_resilience	Flying in windy weather. Runs turbulence stability scoring exclusively. Tests configurations for IMU stability under real gust conditions. Wing simulation also runs in actual turbulence — discovering which failure configurations remain controllable when wind is blowing. Command: disc_mode_engage_focused("gust_resilience")
fault_tolerance	All blocks devoted to wing simulation and progressive failure chains. Maximum fault tolerance data collected per flight hour.
cruise_efficiency	Range per watt-hour at transit lever positions. Best used during a long transit flight where discovery can run alongside the mission.
hover_endurance	Minimum power for SAR loiter. Stability weighted heavily — finds configurations that are both efficient and provide a stable sensor platform.
precision	Minimises sigma_h, sigma_roll, sigma_hdg, sigma_sink. Finds the smoothest, most stable configurations. Valuable before water approach trials.
aerobatic	Maximises aerobatic crispness and hold stability. Tests configurations that produce fast settle times and low sigma_roll during hesitation holds.

SECTION 38 — WING IDLE AND PROGRESSIVE FAILURE SIMULATION

38.1 The Engine-Idle Model

Aviation does not shut down engines to simulate failure — the engine is pulled to idle. The MCO-1D equivalent: a designated wing is set to amplitude zero, swept aft, angle of attack neutral. It stops contributing but is not creating unusual drag. Re-engagement takes one control tick (20ms). The aircraft is never more than one tick from restoring full capability.

38.2 Running a Wing Simulation Test

The pilot must approve every test:

```
wing_sim_request (shows next untested combination) wing_sim_approve (arms the test for next suitable conditions) wing_sim_abort (cancels immediately if needed)
```

Conditions required: h above 50m, turbulence below threshold, airspeed above 1.5× stall speed, no active faults. The test runs for 10 seconds. If altitude drops below 40m, all wings re-engage immediately.

38.3 Progressive Failure Chains

Progressive simulation tests failure sequences: W2p fails first, then while holding the adapted 7-wing configuration, W4s also fails, then W1s. Each step records the best configuration found starting from the already-adapted state of the previous step. 2,080 chains up to 4 steps long are pre-generated, sorted by priority with outermost pairs first.

```
prog_sim_request (shows next priority chain) prog_sim_approve (arms for next flight at 250 feet minimum) prog_sim_status (shows chain test history)
```

SECTION 39 — PRECISION AEROBATICS EXPLORATION

Precision aerobatics requires two qualities that operate at different timescales and appear contradictory. The entry must be crisp — the control input decisive and fast. The hold must be stable — the attitude held precisely until the next input. A conventional aircraft cannot change its aerodynamic damping mid-manoeuvre. The MCO-1D can.

When the roll stick returns to neutral with residual angular velocity, the active damping system immediately applies opposing differential amplitude. A 45°/s residual roll rate is opposed with a 34-degree amplitude differential (port wings at 62°, starboard at 96°) at 50Hz. The aircraft snaps to the current attitude and holds it. The pilot feels: push stick — roll. Release — lock. Push again — roll from locked position. This is the eight-point hesitation roll.

The aerobatic Discovery Mode focus runs the exploration system specifically to find configurations that produce fast settle times (crispness) and low sigma_roll during hold phases (stability). The optimizer discovers which cascade phase and amplitude distribution gives the best combined aerobatic score.

✓ NOTE: All active damping gain constants (ACTIVE_DAMP_K_AMP, ACTIVE_DAMP_K_AOA, ACTIVE_DAMP_SETTLE_TIME_S) require calibration at TC-02 from measured roll response data. The architecture is complete and the gains are estimated conservatively. After TC-02 calibration, aerobatic exploration will be significantly more precise.

PART VIII — ADVANCED SYSTEMS

SECTION 40 — ACTIVE DAMPING SYSTEM

Purpose	Cancel residual angular momentum after a roll input is released. Enables crisp hesitation points in aerobatic manoeuvres.
Trigger	Roll stick returns to within 8% of neutral while roll rate exceeds 2°/s.
Action	Opposing differential amplitude applied at 50Hz. Port and starboard amplitude differ by: roll_rate × ACTIVE_DAMP_K_AMP degrees.
Settle criterion	Roll rate below 2°/s for 3 consecutive ticks.
Hold state	After settling: symmetric amplitude, sigma_roll measurement begins.
K_AMP (current)	0.40 °_amplitude per °/s — estimate. CALIBRATE AT TC-02 from measured roll response.
K_AOA (current)	0.15 °_AoA per °/s — estimate. CALIBRATE AT TC-02.
Settle time est.	0.15 seconds — CALIBRATE AT TC-02.

SECTION 41 — PRECISION DESCENT RATE CONTROLLER

41.1 The FAA Standard

FAA-H-8083-23 specifies approximately 100 to 150 fpm for glassy water and IFR approaches. 100 fpm (0.508 m/s) is the primary target. This is a gentle descent — slower than a lift in a building.

41.2 Physics

The difference between hovering and descending at 100 fpm is approximately 3.6% of hover power. This corresponds to a frequency change of approximately 0.045 Hz — well within the 0.01 Hz command resolution. The controller is feasible. Actual gains require TC-02 calibration.

41.3 Control Law

Height source	Rangefinder primary below 10m. Barometric secondary. Height displayed in feet.
Inner loop	Frequency proportional to descent rate error. Too fast (>target): increase frequency. Too slow: decrease frequency.
Secondary loop	AoA fine trim. Maximum 3° correction.
Outer loop	Height integral prevents long-term drift.
Commands	descent_ctrl_engage 100 (100 fpm target) descent_ctrl_disengage descent_ctrl_status

● CAUTION: Gain constants DESCENT_CTRL_K_FREQ_PER_FPM and DESCENT_CTRL_K_AOA_PER_FPM are estimates pending TC-02. After TC-02, run calibrate_descent_controller(hover_freq_hz=X, lift_freq_slope_N_per_Hz=Y) with measured values.

SECTION 42 — PRECISION SCORING

42.1 What Precision Measures

Every tick during stable flight, the system measures four sigma values from 3-second rolling windows. These are the standard deviations of controlled variables: how much the aircraft drifts from its commanded condition. Lower sigma means tighter control.

sigma_h	Altitude deviation in metres. Excellent below 0.05m. Acceptable below 0.20m.
sigma_roll	Roll angle deviation in degrees.
sigma_hdg	Heading deviation in degrees.
sigma_sink	Descent rate deviation in m/s. Active when descent controller is running.

42.2 Phase-Adaptive Weights

Water approach	sigma_h weighted 0.70 — altitude critical.
SAR loiter	sigma_h 0.50, sigma_hdg 0.20, sigma_roll 0.20.
Transit	sigma_hdg 0.50 — heading drift wastes distance.
Hover	sigma_h 0.50, sigma_roll 0.25.

These weights are applied automatically based on the active flight phase. The pilot does not need to select them.

SECTION 43 — WATER OPERATIONS SOFTWARE

The water operations software framework is complete. Hardware (floats) and hover confirmation (TC-02) are required before use.

Wave estimation	Rangefinder reads water surface height at 50Hz. Significant wave height = 4 × standard deviation. Wave period from zero-crossing rate.
Porpoising detection	IMU pitch rate monitored for 0.3-2.0 Hz oscillation with growing amplitude. If detected: First oscillation: maintain back pressure. Second oscillation: power to idle, hold full elevator back.
Water phase machine	Five states: displacement, plowing, planing, water_approach, airborne. Transitions from sensor data.
Departure timing	Advises departure as float rises toward wave crest. Countdown to optimal departure window in seconds.
Glassy water	100 fpm descent rate. Rangefinder only. No visual altitude reference. Trust instruments.

SECTION 44 — REYNOLDS NUMBER AND BIOLOGICAL SCALING

All 23 biomimetic hypotheses have been corrected for the MCO-1D operating Reynolds number and air medium. This section explains the corrections for reference.

MCO-1D Re range	Re 28,000 in hover to 89,000 in cruise. This is the hawkmoth and small bat regime.
Best biological match	Bat literature (H-02, H-22, H-23): Re 10,000-100,000. Direct overlap. No scaling corrections needed.
AoA corrections	S1223 at Re 50,000: CL_max at 18°, best L/D at 8°. H-04 corrected from 32° to 18°. H-19 corrected from 45° thrust / 82° lift (sea lion in water) to 15° thrust / 22° lift (S1223 in air).
Dimensionless parameters	Phase offsets, amplitude ratios, Strouhal number, advance ratio — all transfer directly across media and Reynolds number. These are geometry ratios, not force values.
Medium (air vs water)	Water is 836× denser than air. Absolute force values from aquatic research do not transfer. H-17 upstroke AoA corrected from 2° (turtle in water with buoyancy) to 5° (MCO-1D in air with no buoyancy).

PART IX — REFERENCE

SECTION 45 — GROUND STATION COMMAND REFERENCE

Complete list of 98 registered commands. Type any command in the ground station command box.

arm / disarm	Arm or disarm ODrives and flight system
emergency_stop	Immediate wing stop, all amplitudes to zero
preflight_start	Begin automated pre-flight checklist
postflight_start	Begin post-flight cooldown and export
ground_ops_taxi	Enter taxi mode for ground movement
storm_stow	Fold wings to storm parking configuration
l9_status	Meta-optimizer status display
exploration_status	Four-layer exploration system status
envelope_status	Demonstrated flight envelope display
margin_status	Current margins on all six axes
knowledge_export	Export discovery registry to file
knowledge_export_usb	Copy knowledge files to USB stick
knowledge_import_usb	Load knowledge from USB stick
knowledge_status	Show current knowledge state
inject_hypothesis	Inject new hypothesis as JSON string
inject_hypothesis_file	Load hypothesis from file path
user_hypothesis_status	Show injected hypotheses
fault_summary	Complete fault and degradation report
failure_briefing	Pre-flight failure performance table
failure_table_status	Failure combination table status
wing_sim_request	Request next wing idle simulation
wing_sim_approve	Approve and arm next simulation test
wing_sim_abort	Abort current simulation
wing_sim_status	Simulation registry status
prog_sim_request	Request next progressive chain
prog_sim_approve	Approve and arm next chain
prog_sim_abort	Abort current chain
prog_sim_status	Progressive chain registry
disc_mode_engage	Engage Discovery Mode (full)

disc_mode_engage_focused	Engage focused Discovery Mode
disc_mode_exit	Exit Discovery Mode
disc_mode_status	Discovery Mode session status
descent_ctrl_engage N	Engage descent controller at N fpm (100 default)
descent_ctrl_disengage	Disengage descent controller
descent_ctrl_status	Descent controller current state
calibrate_descent_ctrl	Apply TC-02 measured gains
precision_status	Current sigma values and precision score
aerobatic_status	Active damping state and aerobatic score
wave_status	Current wave height and period estimate
water_phase_status	Current water operating phase

SECTION 46 — COMPLETE BIOMIMETIC HYPOTHESIS LIBRARY

H-01	Dragonfly	Phase switching: in-phase for force, anti-phase for stability. 5 variants at 0°, 45°, 90°, 135°, 180° phase.
H-02	Bat advance rotation	AoA transitions before stroke reversal. 5 variants at 5-25° advance. Direct Re match.
H-03	Manta spanwise wave	Phase gradient root to tip. 89% propulsive efficiency. 3 variants at 10°, 20°, 30° gradient.
H-04	Pterosaur soar	0.8 Hz, 18° AoA (Re-corrected from 32°), max amplitude. Requires h above 30m.
H-05	Hawkmoth LEV	Spanwise amplitude gradient delays LEV breakdown. 4 variants at 0.9-1.2× tip/root ratio.
H-06	Negative cascade	W4 leads, W1 follows — reversed cascade direction.
H-07	Split frequency	10% inter-pair frequency differential. 3 variants at 5%, 10%, 15%.
H-08	Manta Strouhal	Speed-adaptive: target St 0.20-0.40 optimal range.
H-12	Locust reversal burst	ODrive velocity burst at stroke reversal. 3 variants at 5%, 8%, 12% burst fraction.
H-13	Plesiosaur speed phase	Cascade phase varies with Strouhal number. 4 speed test points.
H-14	Plesiosaur brake	Anti-augmentation: wrong phase = deliberate drag increase.
H-15	Plesiosaur dominance	Fore vs hind pair amplitude ratio. 6 variants from -50% to +100%.
H-16	Sea turtle figure-8	Dynamic in-stroke sweep — new kinematic dimension. 4 variants at 15°, 30°, 45°, 56°.
H-17	Sea turtle asymmetric	Power downstroke, recovery upstroke. 5° AoA floor on upstroke (air medium correction).
H-18	Sea turtle sweep-clap	Sweep-axis clap at bottom of downstroke. 3 variants at 10°, 20°, 35°.
H-19	Sea lion AoA	15° thrust, 22° lift, 0° paddle. (Re-corrected.) 4 force-vector modes.

H-20	Otter advance ratio	Drag-to-lift transition with advance ratio. 5 speed test points.
H-21	Sea lion burst	High-velocity fling at stroke initiation. 3 variants at 1.2×, 1.5×, 2.0× burst.
H-22	Bat ventral clap	Double lift peak from upstroke ventral clap. Direct Re match. 4 variants at 10-35°.
H-23	Bat upstroke AoA	Speed-adaptive: -8° (hover) → +5° (cruise) → -3° (sprint). Direct Re match. 6 speed points.
H-24	Spanwise AoA gradient	MCO-1D original. W1 at thrust AoA, W4 at lift AoA. 4 gradient profiles.
H-25	Per-stroke AoA roll	MCO-1D original. AoA differential for roll, 4× faster than amplitude. 4 variants at 4-16° differential.
H-26	Force vector decoupling	MCO-1D original. Sidestep without banking — entirely unknown until TC-02. 4 variants at 5-20° differential.

SECTION 47 — GPIO PIN REFERENCE

GPIO 4	DS18B20 OAT sensor (One-Wire bus)
GPIO 5	W3p sweep servo
GPIO 6	W3s sweep servo
GPIO 7	HX711 load cell SCK
GPIO 12	W3p pitch servo
GPIO 13	W4p sweep servo
GPIO 16	W3s pitch servo
GPIO 17	W1s sweep servo
GPIO 18	W1p pitch servo
GPIO 19	W4s sweep servo
GPIO 20	W4p pitch servo
GPIO 21	W4s pitch servo
GPIO 22	W2s sweep servo
GPIO 23	W1s pitch servo
GPIO 24	W2p pitch servo
GPIO 25	W2s pitch servo / MCP2515 CAN interrupt
GPIO 26	HX711 load cell DT (data)
GPIO 27	W2p sweep servo
SPI0	MCP2515 CAN-FD adapter (MOSI, MISO, SCLK, CE0)
I2C1	IMU (MPU-6050) and barometer (BMP280)
UART	Pixhawk MAVLink at 921600 baud

SECTION 48 — SOFTWARE FILE REFERENCE

mco_wave_library.py	42,115 lines. Main flight control library. Contains all optimizer, hypotheses, fault tolerance, and advanced systems.
mco_main.py	307 lines. Flight loop entry point. Reads sensors, calls tick(), sends commands.
mco_mavlink.py	290 lines. Pixhawk MAVLink bridge. Provides GPS, compass, extra IMU.
mco_gs_server.py	193 lines. TCP socket server for ground station.
mco_logger.py	185 lines. 50 Hz CSV flight data logger.
mco_odrive_setup.py	187 lines. ODrive configuration and diagnostics.
mco_panel.py	485 lines. 4-button touchscreen + margin bar.
mco_ground_station.py	588 lines. Laptop ground station client.
mco_sync.py	229 lines. Knowledge transfer between Pis.
mco_hypothesis_tool.py	287 lines. User hypothesis authoring and injection.
mco_loadcell.py	TO BE WRITTEN. HX711 reader for TC-02 thrust.
mco_can_setup.sh	CAN interface shell script.
mco_install.sh	One-step installation script.
mco-flight.service	Systemd auto-start service definition.
mco-panel.service	Systemd panel auto-start.

SECTION 49 — STATISTICS AND PENDING ITEMS

49.1 Software Statistics

Wave Library lines	42,115
Total Python files	10 (plus mco_loadcell.py pending)
Tick stack depth	53 layered wrappers + base tick
Control loop rate	50 Hz (20ms per tick)
Active hypotheses	23 (H-01 to H-26, excluding H-09/10/11)
Pending hypotheses	3 (H-09, H-10, H-11 — bench validation needed)
Hypothesis variants	93 total across all active hypotheses
Biological sources	11 animals (dragonfly to otter)
Failure combinations	255 pre-computed
Progressive chains	2,080 generated
Discovery focus modes	10
Ground station commands	98
SD card write I/O	Zero in tick loop — all writes in post-flight
cl_cd cache hit rate	96%+ after warm-up (first 182 ticks)
Tick time dev machine	0.85ms mean, 3.1ms P99
Tick time Pi 4 estimate	~10ms mean (2× headroom vs 20ms budget)

49.2 Pending Items Before TC-01

Build Measurement Sheet	Fill all physical dimensions from assembled hardware. Update CHORD_MM, SEMI_SPANS_MM, and geometry constants.
ODrive vel_limit	Set to 150 rad/s via mco_odrive_setup.py. Verify saved.
RC channel mapping	Verify CH1=roll, CH2=pitch, CH3=lever, CH4=yaw.
mco_loadcell.py	Write HX711 load cell reader for TC-01 static thrust measurement.

49.3 Pending Items After TC-02

Descent controller calibration	Run calibrate_descent_controller() with measured hover frequency and lift-frequency slope.
Active damping calibration	Set ACTIVE_DAMP_K_AMP and ACTIVE_DAMP_K_AOA from measured roll rate response.

H-09, H-10, H-11	H-09 needs ODrive PWM validation. H-10 needs bench rope drive test. H-11 needs sweep servo speed measurement.
H-26 behaviour	Is force vector decoupling (sidestep) real? Or does AoA differential simply cause roll? Answer only from TC-02 flight test.

49.4 Post TC-03 Development

Stroke plane controller	Generates complete 8-wing commands from a desired world-frame force vector. Enables backward flight, sideways translation, knife-edge split stroke. Requires measured CT-J curve from TC-03.
GPS position controller	Improves Discovery Mode racetrack precision. Currently heading-plus-time only.
Float specification	Water operations hardware BOM. Requires hover confirmation first.
Water operations procedures	Cannot finalise until float geometry and hover data are known.